Regras de derivação

Sendo f(x) e g(x) funções e c uma constante, temos $u = f(x)$ e $v = g(x)$, então :

$y = c$	$y' = 0$
$y = u^n$	$y' = n.u^{n-1}$
$y = u + v$	$y' = u' + v'$
$y = u.v$	$y' = u'.v + u.v'$
$y = \frac{u}{v}, v \neq 0$	$y' = \frac{u'.v - u.v'}{v^2}$
$y = \text{sen } u$	$y' = u' \cos u$
$y = \cos u$	$y' = -u' \text{sen } u$
$y = \text{tg } u$	$y' = u' \sec^2 u$
$y = \sec u$	$y' = u' \sec u.\text{tg } u$
$y = \text{cosec } u$	$y' = -u' \text{cossec } u.\text{cotg } u$
$y = \cot g \text{ u}$	$y' = -u' \text{cossec}^2 u$
$y = a^u$	$y' = u' a^u \ln a$
$y = e^u$	$y' = u' e^u$
$y = \log_a u$	$y' = \frac{u'}{u} \log_a e$
$y = \ln u$	$y' = \frac{u'}{u}$
$y = \text{arc sen } u$	$y' = \frac{u'}{\sqrt{1 - u^2}}$
$y = \text{arc cos } u$	$y' = \frac{-u'}{\sqrt{1 - u^2}}$
$y = \text{arc tg } u$	$y' = \frac{u'}{1 + u^2}$